Shelter Animal Outcome Prediction

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https://github.com/AnnaW0209/final_project_animal_adoption

Introduction

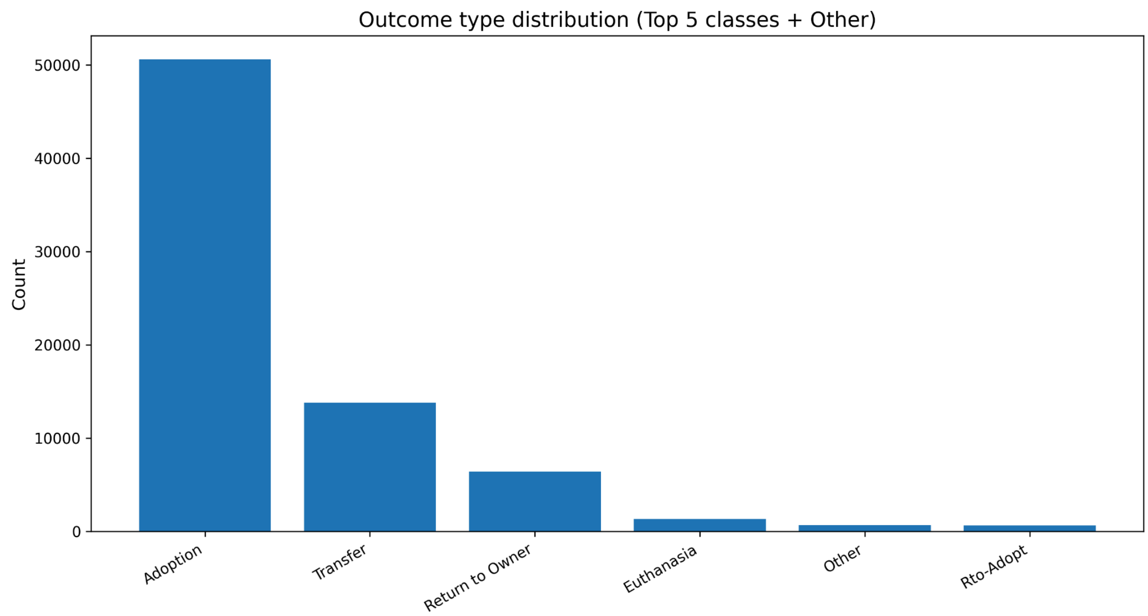
Animal shelters regularly face capacity constraints and must make rapid decisions about the intake and placement of animals. Accurately predicting the likely outcome of an animal upon intake, such as adoption, transfer, or return to owner, can help shelters to allocate resources more efficiently, prioritize high-risk cases, improve adoption, and enhance overall animal welfare. In this project, I formulate this task as a multi-class classification problem, aiming to predict the first outcome upon intake for animals entering a municipal shelter.

I use publicly available data from the **Austin Animal Center**,[1] which provides detailed records of animal intakes and outcomes over multiple years. Each record contains demographic information including animal type, breed, sex, color, intake conditions, have a name or not, and temporal features derived from intake timestamps. The target variable consists of multiple outcome categories, and the top four are: Adoption, Transfer, Return to Owner, and Euthanasia. The dataset is inherently imbalanced, with adoption-related outcomes dominating the class distribution, thus rendering naive accuracy an inappropriate evaluation metric.

The same prediction tasks [2][3] have been explored in prior community-driven machine learning projects using the Austin Animal Center dataset. These works typically train standard classifiers such as logistic regression, random forests, or gradient boosting models on intake-level features and report only moderate predictive performance, particularly when evaluated using class-balanced metrics. This limited performance is largely attributed to serve strong class imbalance, substantial overlap between outcome categories, and the inherent uncertainty of predicting shelter outcomes from intake information alone. Accordingly, the predictive performance achieved in this project is expected to be comparable to previous work, and substantially higher performance would likely indicate issues such as data leakage or improper evaluation.

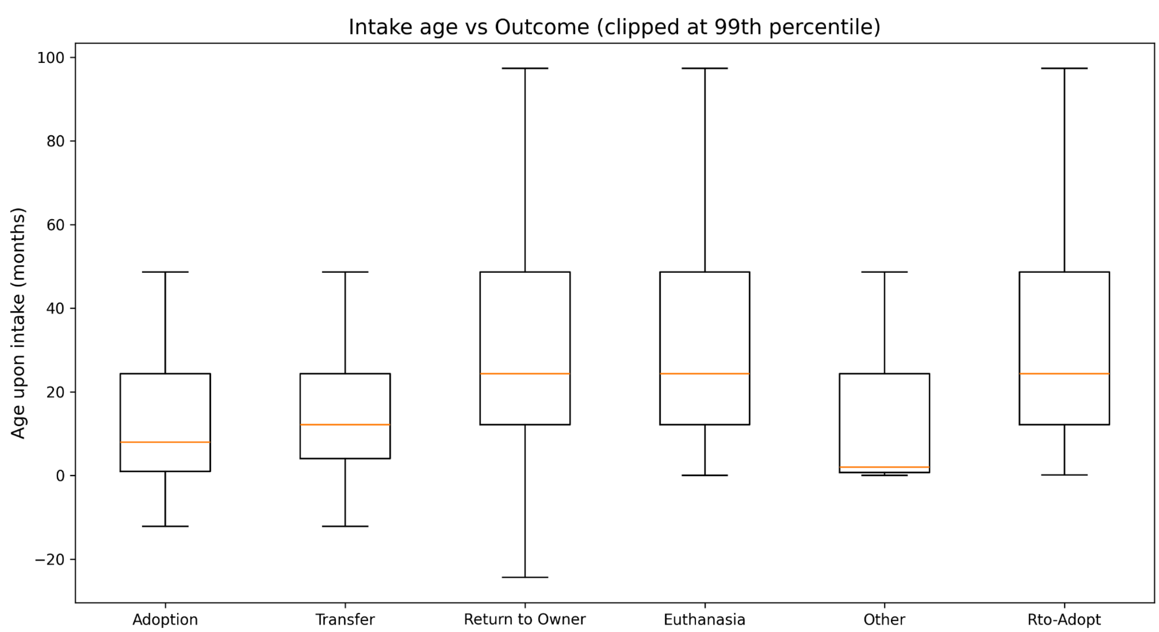
EDA

I conducted EDA to understand the distribution of outcomes and the relationships between key intake features and animal outcomes. The dataset exhibits class imbalance, with Adoption outcomes primarily dominating the distribution. By analyzing the outcomes distribution, I selected the top four outcomes and classified all the other minor outcomes to ‘Other’ to reduce noise and improve model stability. This imbalance motivates the use of evaluation metrics that treat all classes equally and informs the modeling choices discussed later.

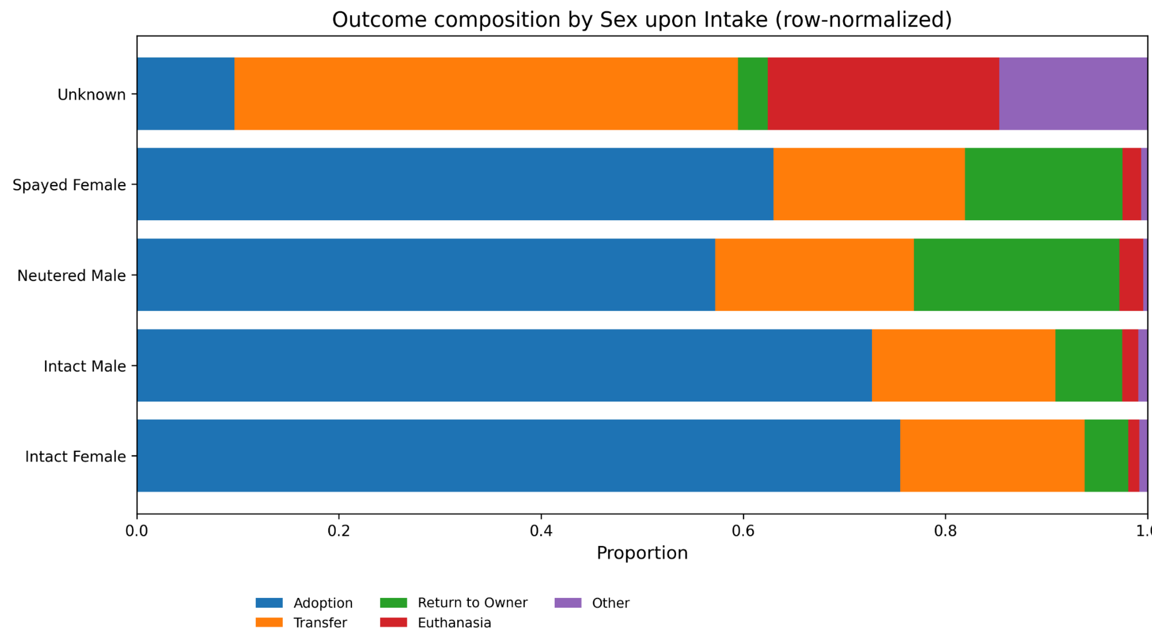


<Graph 1: Outcome Types Distribution Bar Plot>

Several intake features exhibit clear associations with outcomes. Animal type, sex, and age at intake show systematic differences across outcome categories; for example, younger and neutered animals are more likely to experience favorable outcomes. Temporal patterns are also observed, as outcome frequencies vary across months, suggesting seasonal effects in shelter operations.

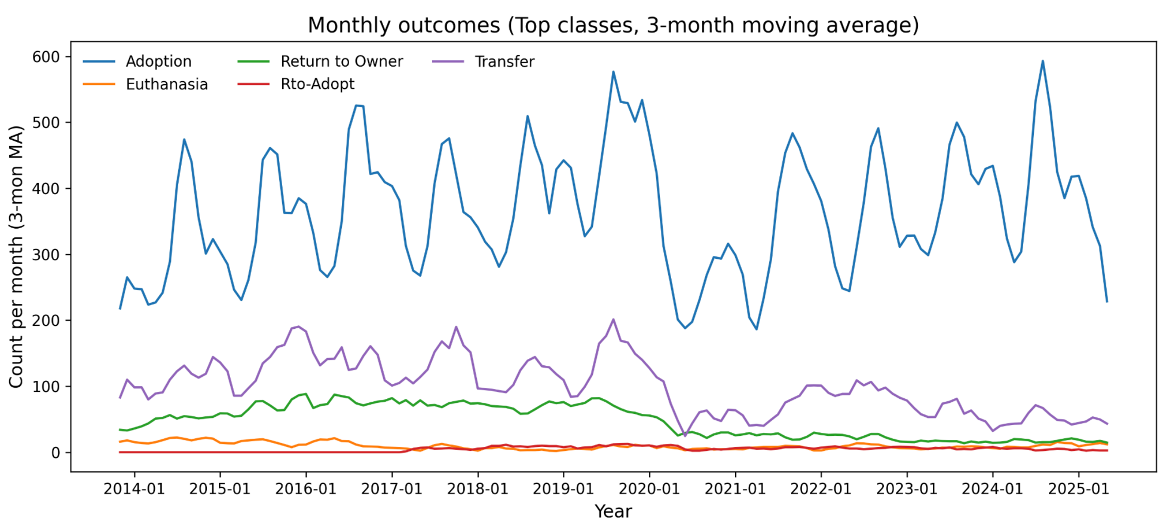


<Graph 2: Intake Age vs Outcome Box Plot>



<Graph 3: Outcome composition by Sex Stacked Bar Plot>

Despite these patterns, substantial overlap exists between outcome categories across most features. This overlap highlights the intrinsic difficulty of the classification task and indicates that intake-level information alone may be insufficient to perfectly separate rare outcomes.



<Graph 4: Monthly Outcomes Line Chart>

Methods

Data Preparation and Label Construction

The intake and outcome records were provided as two separate tables. To construct a meaningful prediction target, intake records were matched to outcome records using animal IDs and timestamps, ensuring that each intake was paired with its corresponding first recorded outcome. This step avoids information leakage from future outcomes and ensures temporal consistency in the prediction task. Missing value analysis revealed that most features were complete. The only feature with missing values was the animal name. Rather than imputing names directly, I converted this feature into a binary indicator representing whether an animal has a recorded name (has_name). This transformation preserves useful information while avoiding arbitrary imputation. No blank values were observed in the dataset, and a small number of “unknown” entries appeared as valid string categories in the original data. I still included a simple imputation in the preprocessing pipeline for completeness, but it did not materially affect the dataset.

Data Splitting Strategy

To prevent data leakage across repeated records of the same animal, all data splitting was performed using a group-aware strategy, where all records associated with the same animal were assigned to either the training or test set, but never both. The dataset was first split into training and test sets using a group shuffle split with an 80/20 ratio. To reduce computational cost during model selection, I randomly drew a subsample of size 1000 still based on groups from the training set. This subsample preserves the group structure and class distribution and was used exclusively for hyperparameter tuning and model comparison. The final evaluation was always performed on the full held-out test set. For the XGBoost model, early stopping was employed to prevent overfitting. Specifically, the training set was further split into training and validation subsets using a group shuffle split again with a validation size of 20%. The number of boosting rounds was determined based on the validation performance, and the best n_estimators were retained at about 116.

Preprocessing and Feature Engineering

Categorical features were encoded using one-hot encoding, while numerical features were standardized. All preprocessing steps were implemented within a unified pipeline to ensure consistent transformations across training, validation, and test sets.

Models and Hyperparameter Tuning

I used four machine learning models, linear and non-linear: Logistic regression, Decision Tree, Random Forest, and XGBoost. I also considered SVM and KNN, but excluded them due to the high computational cost associated with the 399-column feature space after one-hot encoding. Below are the hyperparameters tuned. (Table 1)

Model	Type	Parameters Tuned
Logistic Regression	Linear	C: [logspace(-3,3,7)] penalty: [l2]
Decision Tree	Non-linear	max_depth: [None,3,5,10] min_samples_split: [2, 10, 50] min_samples_leaf: [1,5,10] max_features: [None, ‘sqrt’, ‘log2’]

Random Forest	Non-linear (tree-based)	n_estimators: [20,100,200,500] max_depth: [None, 5, 10, 20] min_samples_split: [2,5,10,20]
XGBoost	Non-linear (boosted trees)	max_depth: [1,5,10,30] learning_rate: [0.01,0.05,0.1] max_subsample: [0.3,0.5,0.7,0.9] reg_lambda: [0.01,1,100]

<Table 1. Tuned parameter values for each machine learning algorithm>

Evaluation Metric and Uncertainty Analysis

Given this is an imbalance multi-class classification problem, model performance was evaluated using the macro-averaged F1 score, which treats all outcome categories equally and is therefore more appropriate than accuracy. To quantify uncertainty due to data splitting, the final selected models were evaluated using repeated group-based shuffle splits. For each model, 20 different train–test splits were generated, and the macro-F1 score was computed for each split. The resulting distribution of scores was summarized using the mean and standard deviation, providing an estimate of test-time performance variability.

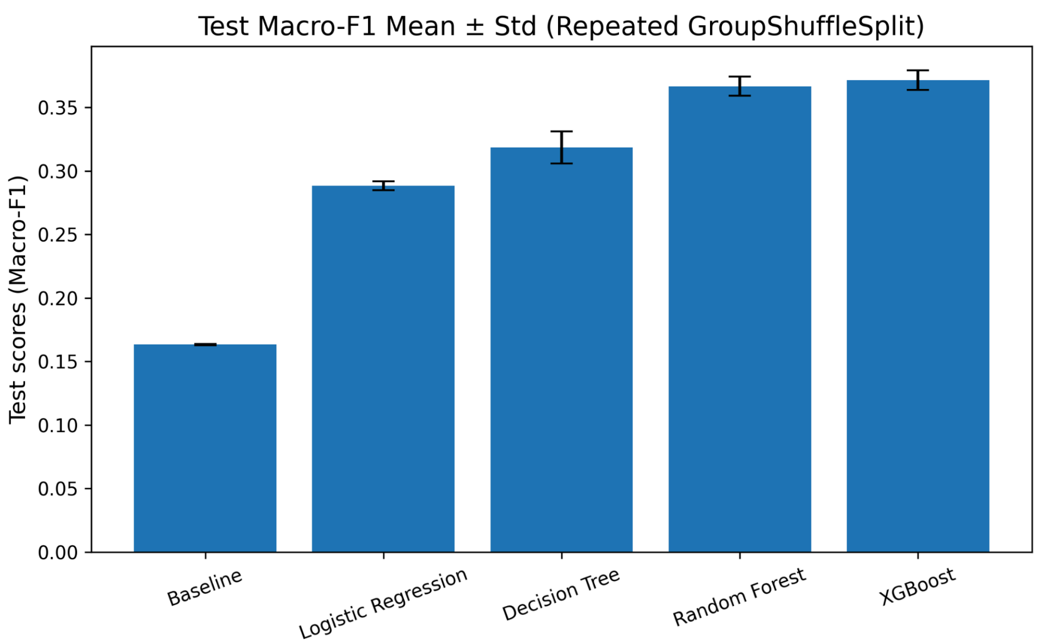
Results

Model performance was evaluated using the macro-averaged F1 score and compared against a dummy classifier baseline that predicts outcomes according to class frequencies. All machine learning models outperform the baseline, indicating that intake-level features contain meaningful predictive signals despite strong class imbalance. Below is the table of test scores made by all models. (Table 2) Where the sigma_above_baseline is calculated as the difference between the mean test macro-F1 of the model and the baseline, normalized by the baseline's test-time standard deviation. Among the evaluated models, XGBoost achieved the highest mean macro-F1 score on the held-out test set, improving the F1 score from baseline ~0.16 to ~0.37. This was closely followed by the Random Forest, which also achieved a ~0.36 test score and even 0.0001 lower standard deviation. Linear model - logistic regression shows lower predictive performance, reflecting their limited capacity to capture non-linear interactions among intake features. Overall, performance differences across models are moderate, consistent with previous work on the same dataset.

	model	test_accuracy	test_recall	test_macro_f1	test_mean_macro_f1	test_std_macro_f1	test_macro_fbeta2	sigma_above_baseline
0	Baseline	0.686516	0.200000	0.162825	0.163255	0.000479	0.183263	0.00

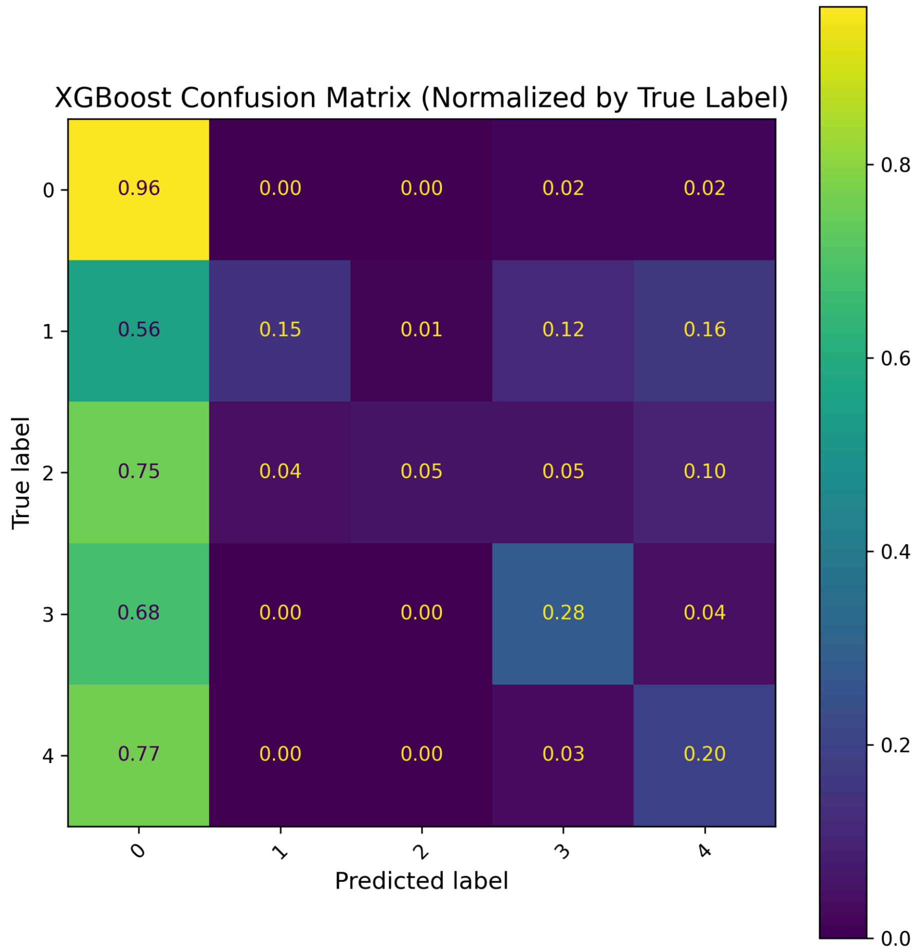
1	Logistic Regression	0.457110	0.386891	0.289467	0.288255	0.003497	0.318387	35.42
2	Decision Tree	0.714587	0.291973	0.315486	0.318403	0.012597	0.294878	12.31
3	Random Forest	0.707570	0.327465	0.362638	0.366584	0.007607	0.336266	26.68
4	XGBoost	0.726783	0.328775	0.369641	0.371353	0.007769	0.337947	26.73

<Table 2: Test Scores of Each Machine Learning Model>



<Graph 5: Test Scores of All Models Bar Plot>

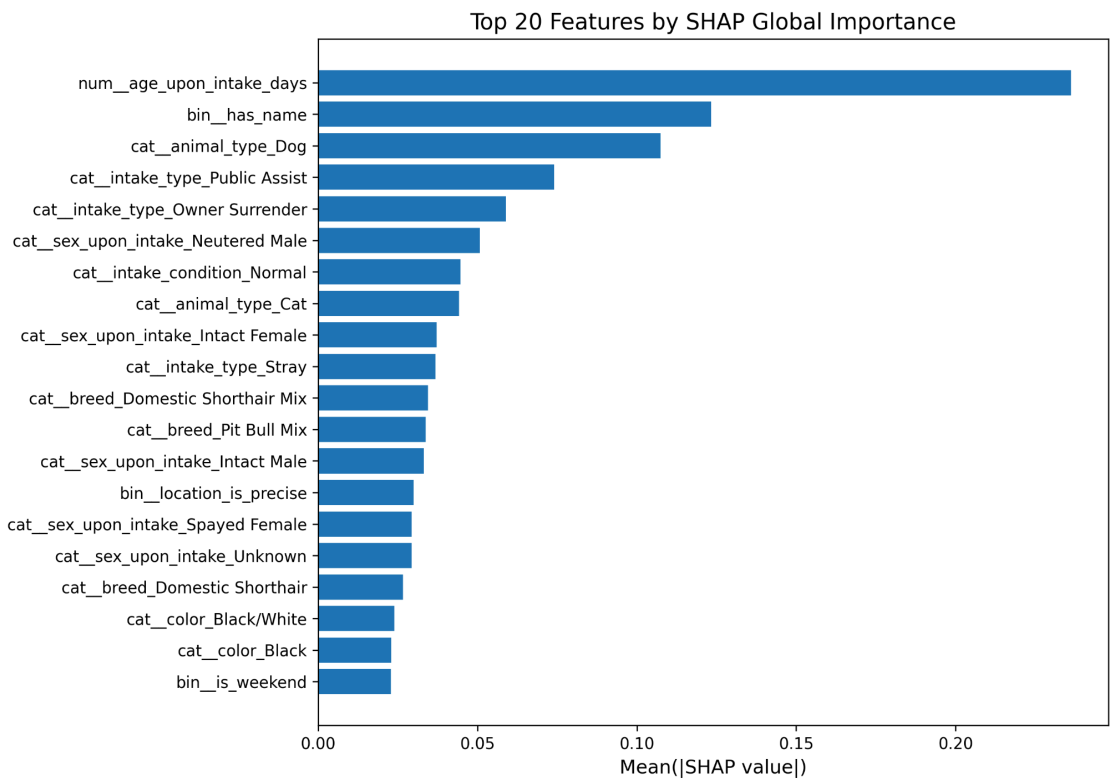
Confusion Matrix
The confusion matrix of the best XGBoost model reveals systematic patterns in prediction errors: Adoption outcomes are predicted with high accuracy, while misclassifications primarily occur among less frequent outcome categories. Rare outcomes are often confused with more common classes, reflecting both class imbalance and overlapping feature distributions. These patterns suggest that model errors are driven by intrinsic data limitations rather than overfitting or evaluation artifacts.



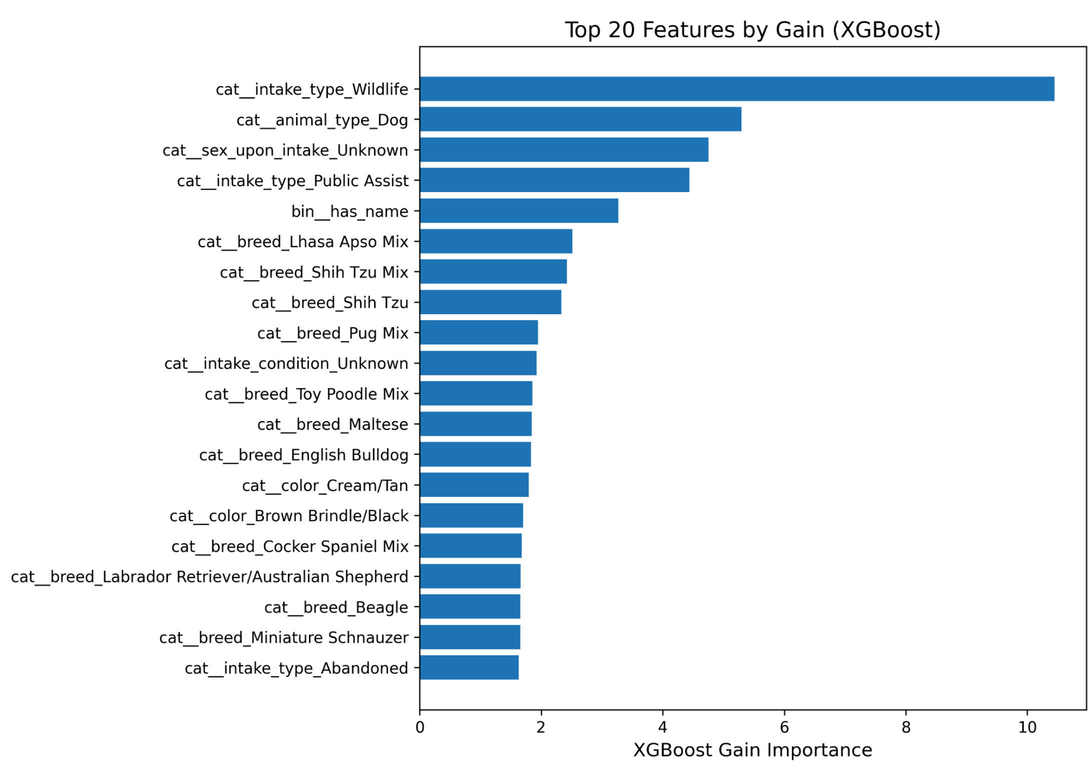
<Graph 6: XGBoost Confusion Matrix Plot>

Feature Importance and Interpretability

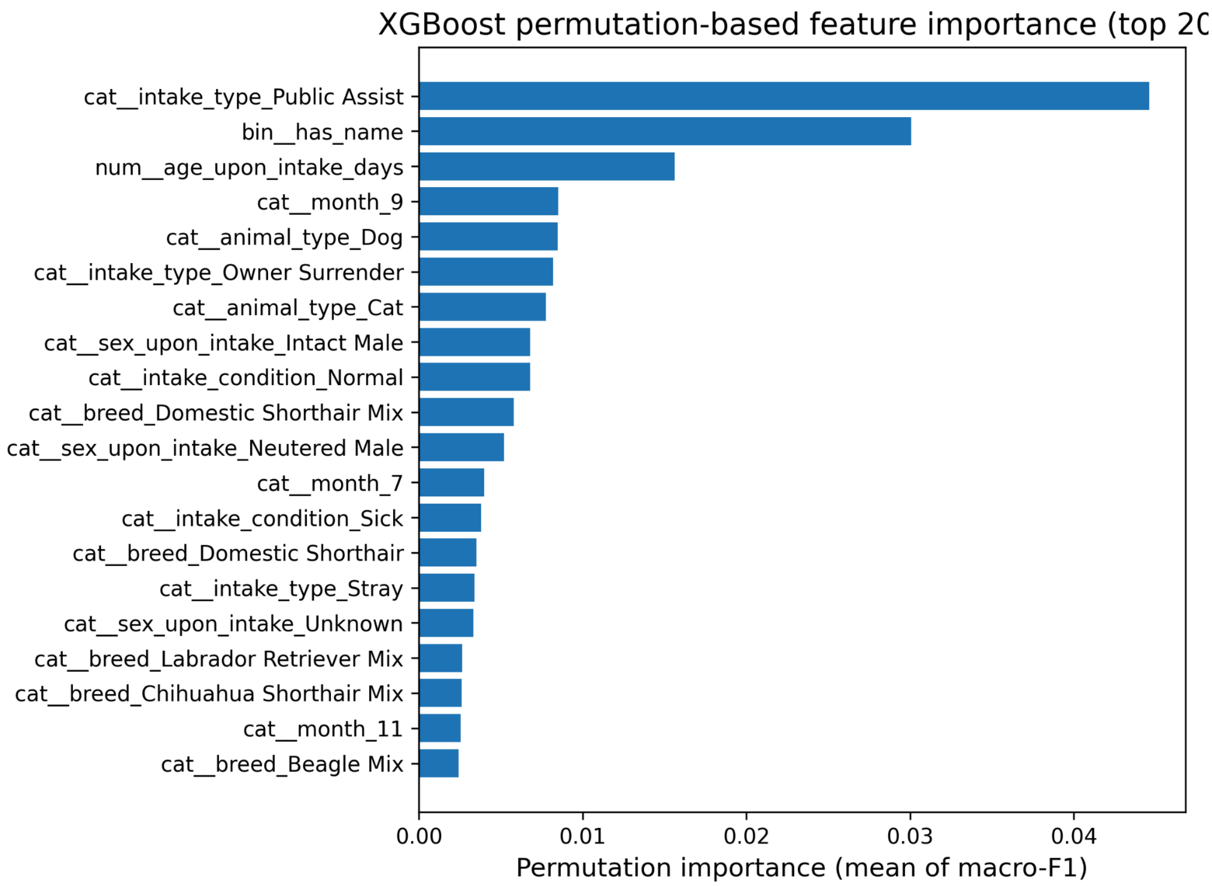
Feature importance was analyzed using multiple global importance measures, including model-based importance scores and permutation-based methods. Across all approaches, age at intake, animal type, sex, and intake condition consistently rank among the most influential features, indicating robust relevance. It's surprising that whether the animal has a name or not is highly ranked among all three approaches and actually plays an important role in the outcome. While the name itself does not causally influence outcomes, this feature likely serves as a proxy for prior human interaction and socialization. It makes sense as animals that already have names are more likely to have been previously owned or cared for, which increases their likelihood of adoption or return to owner. This result highlights how simple intake-level indicators can capture meaningful latent information not explicitly encoded in other features.



<Graph 7: Global SHAP Importances (Top 20)>

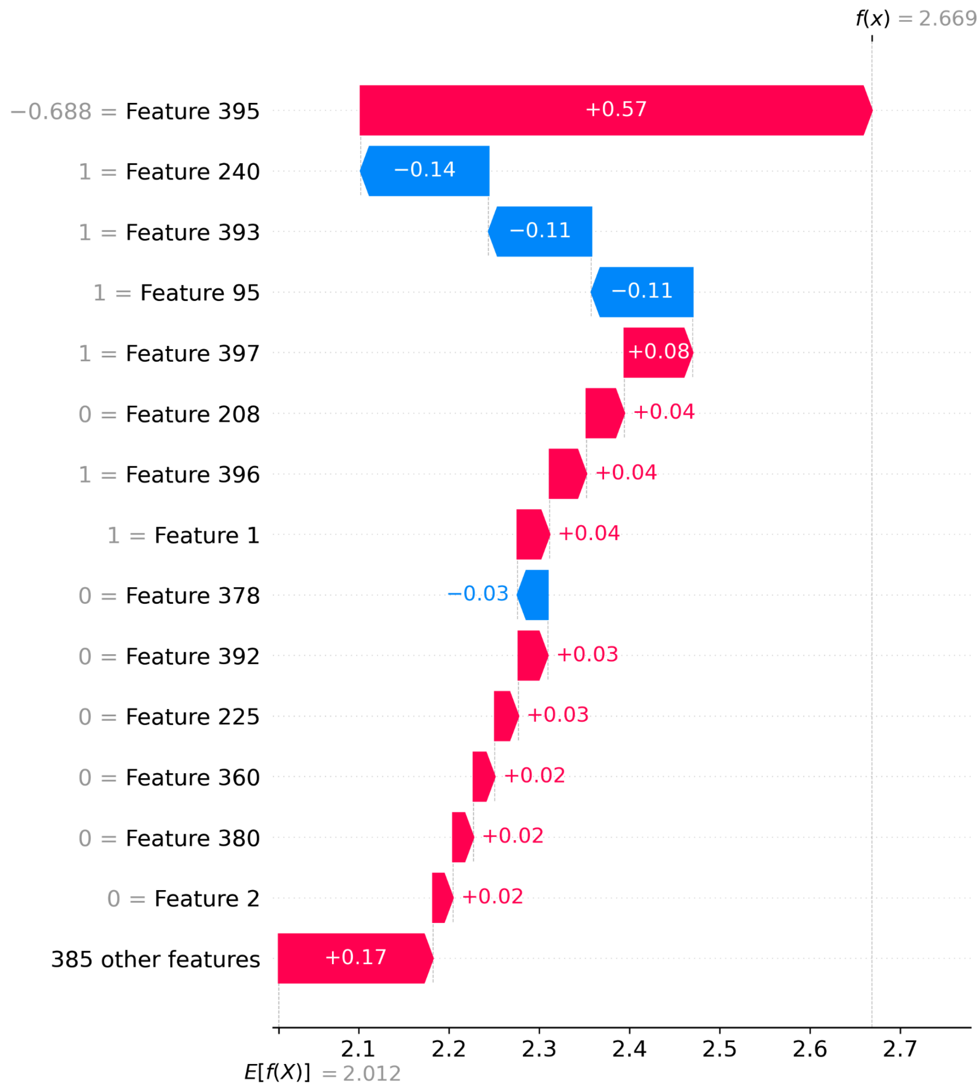


<Graph 8: XGBoost Built-in Feature Importance (Gain)>



<Graph 9: Permutation Importance>

Local interpretability was further examined using SHAP values for individual predictions.[4] I randomly evaluated at the 209th sample, which has a true outcome of "Adoption", and Model prediction prob: [0.89167696 0.0077131 0.01841777 0.00618564 0.0760065] with the Pred class index: 0. SHAP analysis confirms that younger age and favorable intake conditions contribute positively toward adoption-related outcomes, while older age and adverse intake conditions shift predictions toward less favorable outcomes. These local explanations align with domain expectations and support the global importance findings.



<Graph 10: Local SHAP at 209th Sample>

Outlook

While the developed models demonstrate meaningful predictive power, several limitations remain.

First, the prediction task relies exclusively on intake-level information, whereas many outcomes are influenced by post-intake factors such as shelter capacity, behavioral assessments, and medical interventions that are not captured in the dataset. Incorporating such information could improve the prediction of rare outcomes. One example feature can be used from this data is Length of Stay. However, since the lengths are subject to change with the outcome date, incorporating it would require jointly modeling intake and outcome timelines, and might be better to predict the final outcome rather than the first.

Second, class imbalance remains a fundamental challenge. Although the macro-F1 metric helps to mitigate some bias, future work could apply class-weighted functions or resampling techniques to further improve minority class performance.

Finally, interpretability could be extended by aggregating local explanations across subpopulations or by integrating domain-specific constraints.
These directions can help further translate predictive models into practical tools for shelter operations.

References

[1] Austin Animal Center. Austin Animal Center Intake and Outcome Data. https://data.austintexas.gov/
[2] Kaggle. Starter: Austin Animal Center Shelter Outcomes.
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[4] Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems.